Programming Languages and Computation

Week 7: Operational Semantics & Hoare Logic

1 Operational Semantics

This section is about the small-step operational semantics of While programs as given by the relation $\rightarrow \subseteq \mathcal{C} \times \mathcal{C}$ where the configurations are either a statement and a state or a state $\mathcal{C} = (\mathsf{S} \times \mathsf{State}) \cup \mathsf{State}$, which is defined by these following rules:

$$\frac{}{\langle \mathsf{skip}, \sigma\rangle \rightarrow \sigma} \qquad \frac{}{\langle x \leftarrow e, \sigma\rangle \rightarrow \sigma[x \mapsto [\![e]\!]_{\mathcal{A}}(\sigma)]}$$

$$\frac{\langle S_1, \sigma_1\rangle \rightarrow \sigma_2}{\langle S_1;\, S_2, \sigma_1\rangle \rightarrow \langle S_2, \sigma_2\rangle} \qquad \frac{\langle S_1, \sigma_1\rangle \rightarrow \langle S_1', \sigma_2\rangle}{\langle S_1;\, S_2, \sigma_1\rangle \rightarrow \langle S_1';\, S_2, \sigma_2\rangle}$$

$$\frac{}{\langle \mathsf{if}\ e\ \mathsf{then}\ S_1\ \mathsf{else}\ S_2, \sigma\rangle \rightarrow \langle S_1, \sigma\rangle}\ [\![e]\!]_{\mathcal{B}}(\sigma) = \top$$

$$\frac{}{\langle \mathsf{if}\ e\ \mathsf{then}\ S_1\ \mathsf{else}\ S_2, \sigma\rangle \rightarrow \langle S_2, \sigma\rangle}\ [\![e]\!]_{\mathcal{B}}(\sigma) = \bot$$

$$\frac{}{\langle \mathsf{while}\ e\ \mathsf{do}\ S, \sigma\rangle \rightarrow \sigma}\ [\![e]\!]_{\mathcal{B}}(\sigma) = \bot$$

$$\frac{}{\langle \mathsf{while}\ e\ \mathsf{do}\ S, \sigma\rangle \rightarrow \langle S;\, \mathsf{while}\ e\ \mathsf{do}\ S, \sigma\rangle}\ [\![e]\!]_{\mathcal{B}}(\sigma) = \top$$

Figure 1: Operational semantics for While.

* 1. Calculate the terminal state and write down a trace for the statement $x \leftarrow 1; \{x \leftarrow 2; x \leftarrow 3\}$ with the initial state $[]$ using the rules in Figure 1. Remember, variables are assigned 0 by default.

* 2. Calculate the terminal state and write down a trace for the statement $\{x \leftarrow 1; x \leftarrow x*2\}; x \leftarrow x+y$ with the initial state $[x \mapsto 2,\, y \mapsto 2]$ using the rules in Figure 1.

* 3. Find a state σ such that $\langle x \leftarrow 1; y \leftarrow x*2, []\rangle \rightarrow^* \sigma$. You should provide the corresponding trace.

* 4. Compute the final state for the program if $x \leq y$ then $x \leftarrow y$ else $y \leftarrow x$ when executed in each of the following states:

- $[]$
- $[x \mapsto 2, y \mapsto 3]$
- $[x \mapsto 4, y \mapsto 2]$

* 5. Find a state $\sigma \in$ State such that while $!(x \leq -1)$ do $x \leftarrow x + d, [d \mapsto -1] \rightarrow^* \sigma$. You should provide the corresponding trace.

* 6. Find a state $\sigma \in$ State such that $\langle x \leftarrow 2; y \leftarrow x * y, \sigma\rangle \rightarrow^* [x \mapsto 2, y \mapsto 4]$. You should provide the corresponding trace.

** 7. Suppose $e \in \mathcal{B}$ is a Boolean expression that is semantically equivalent to false. Prove that $\langle$while e do $S, \sigma\rangle \rightarrow \sigma$ for any state $\sigma \in$ State.

** 8. Suppose $S_1, S_2 \in \mathcal{S}$ are two statements such that $\langle S_1, \sigma\rangle \rightarrow^* \sigma'$ and $\langle S_2, \sigma\rangle \rightarrow^* \sigma'$ for some states $\sigma, \sigma' \in$ State. Prove that if e then S_1 else $S_2, \sigma \rightarrow^* \sigma'$ for any Boolean expression $e \in \mathcal{B}$.

** 9. Suppose we introduce a new language construct "do S while e" where $S \in \mathcal{S}$ is a statement and $e \in \mathcal{B}$ is a Boolean expression. The operational semantics for this construct is given by the following inference rules:

$$\frac{}{\langle \text{do } S \text{ while } e, \sigma\rangle \rightarrow \langle S;\ \text{if } e \text{ then do } S \text{ while } e \text{ else skip}, \sigma\rangle}$$

(a) Find a state $\sigma \in$ State such that $\langle$do $x \leftarrow x+1$ while $x \leq 1, []\rangle \rightarrow^* \sigma$ and give the associated trace.

(b) For a given statement $S \in \mathcal{S}$ and a Boolean expression $e \in \mathcal{B}$ find a While program that is equivalent to the program do S while e but does not use the new construct. That is, find a statement $S' \in \mathcal{S}$ such that:

$$\langle S', \sigma\rangle \rightarrow^* \sigma' \Leftrightarrow \langle \text{do } S \text{ while } e, \sigma\rangle \rightarrow^* \sigma'$$

You do not need to prove that your answer is correct but should provide a trace for $\langle S', []\rangle \rightarrow^* \sigma$ where S is given to be the statement $x \leftarrow x + 1$, where e is given to be the expression $x \leq 1$, and where σ is the state from part (a).

** 10. Suppose we introduce a new language construct for x do S where $S \in \mathcal{S}$ is a statement and $x \in$ Var is a variable The operational semantics for this construct is given by the following inference rules:

$$\frac{}{\langle \text{for } x \text{ do } S, \sigma\rangle \rightarrow \sigma}\ \sigma(x) \leq 0 \qquad \frac{}{\langle \text{for } x \text{ do } S, \sigma_1\rangle \rightarrow \langle S;\ \text{for } x \text{ do } S, \sigma[x \mapsto \sigma(x) - 1]\rangle}\ \sigma(x) > 0$$

(a) Find a state $\sigma \in \mathsf{State}$ such that $\langle \mathsf{for}\ x\ \mathsf{do}\ y \leftarrow y + x;\ x \leftarrow x - 2, [x \mapsto 3]\rangle \to^* \sigma$ and give the associated trace.

(b) For a given statement $S \in \mathcal{S}$ and a variable $x \in \mathsf{Var}$ find a While program that is equivalent to the program for x do S but does not use the new construct. That is, find a statement $S' \in \mathcal{S}$ such that:

$$\langle S', \sigma\rangle \to^* \sigma' \Leftrightarrow \langle \mathsf{for}\ x\ \mathsf{do}\ S, \sigma\rangle \to^* \sigma'$$

You do not need to prove that your answer is correct but should provide a trace for $\langle S', [x \mapsto 3]\rangle \to^* \sigma$ where S is given to be the statement $y \leftarrow y + x;\ x \leftarrow x - 2$ and σ is the state from part (a).

*** 11. Show that, if $y \notin \mathsf{FV}(e_1)$ and $x \notin \mathsf{FV}(e_2)$, then the statements $x \leftarrow e_1;\ y \leftarrow e_2$ and $x \leftarrow e_1;\ y \leftarrow e_2$ equivalent in the sense that $\langle x \leftarrow e_1;\ y \leftarrow e_2, \sigma\rangle \to^* \sigma'$ if and only if $\langle y \leftarrow e_2;\ x \leftarrow e_1, \sigma\rangle \to^* \sigma'$. You may use results from the previous worksheet.

*** 12. Suppose σ_1 and σ_1' are two states and S is some statement such that $\sigma_1(x) = \sigma_1'(x)$ for all $x \in \mathsf{FV}(S)$. Prove that if $\langle S, \sigma_1\rangle \to \langle S', \sigma_2\rangle$ for some statement S' and some state σ_2, then $\langle S, \sigma_1'\rangle \to \langle S', \sigma_2'\rangle$ for some state σ_2' such that $\sigma_2(x) = \sigma_2'(x)$ for all $x \in \mathsf{FV}(S)$.

Using this result, show that if $\langle S, \sigma_1\rangle \to^* \sigma_2$, then $\langle S, \sigma_1'\rangle \to^* \sigma_2'$ for some states σ_2 and σ_2' such that $\sigma_2(x) = \sigma_2'(x)$ for all $x \in \mathsf{FV}(S)$ where the same assumption is made about σ_1 and σ_1'.

Hoare Logic

* 13. Consider the following *invalid* Hoare triple: $\{x \le y\}\ y \leftarrow y * 2\ \{x \le y\}$. Find an initial state $\sigma \in \mathsf{State}$ and a trace from the configuration $\langle y \leftarrow y * 2, \sigma\rangle$ that contradicts this triple.

* 14. Find a statement S such that $\{\mathsf{true}\}\ S\ \{z \le x\ \&\&\ z \le y\}$.

** 15. For each of the following statements, compute the strongest post-condition from the pre-condition $(\exists q. x = q * y)\ \&\&\ y \ge 0$:

(a) $x \leftarrow x * x$

(b) $\mathsf{if}\ y \le x\ \mathsf{then}\ x \leftarrow x - y\ \mathsf{else}\ y \leftarrow y - x$

** 16. Using your answers to the previous question, for each of the following Hoare triples determine whether they are valid or not. You should justify your answer.

(a) $\{(\exists q. x = q * y)\ \&\&\ y \ge 0\}\ x \leftarrow x * x\ \{(\exists q. x = q * y)\}$

(b) $\{(\exists q.\, x = q * y) \mathbin{\&\&} y \geq 0\}\ x \leftarrow x * x\ \{x \geq y\}$

(c) $\{(\exists q.\, x = q * y) \mathbin{\&\&} y \geq 0\}$ if $y \leq x$ then $x \leftarrow x - y$ else $y \leftarrow y - x$ $\{(\exists q.\, x = q * y) \mathbin{\&\&} y \geq 0\}$

** 17. Something problems involving the consequence rule.

*** 18. Some questions that explore *partial* correctness.

*** 19. The rule of consequence allows us to weaken a Hoare triple for use in a different context. However, consider when S is some (unknown) statement with the triple $\{\mathrm{T}\}\ S\ \{z \geq x \mathbin{\&\&} z \geq y\}$. But it doesn't tell us anything about whether S effects some other variable w.